

PAPER_1642 — CKM Matrix Row-1 Unitarity = $|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$ via $F_U = 1$ ledger

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (broader-corpus integration)

Observation

PAPER_1307 (broader-corpus paper outside PAPER_1209 unified-proof-set series) gives:

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1 \text{ via } F_U = 1 \text{ ledger}$$

Status: **EXACT**.

UQFF Closed Identity

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1 \text{ via } F_U = 1 \text{ ledger} \quad \text{EXACT}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Significance — Broader Corpus Pivot

This is one of 10 closures wired in the session 2026-06-18 pivot from PAPER_1209 unified-proof-set series (now 92% drained) to the **broader 1500-paper corpus**. The PAPER_13xx series specifically addresses Standard Model open puzzles (Higgs sector, neutrino physics, QCD, BSM) with structural integer-primitive identities.

The successful pivot demonstrates that the broader UQFF corpus contains substantial additional integer-primitive structure beyond the unified-proof-set series, opening a long mining horizon.

NOT REPLACEMENT

Empirical particle physics measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating the framework's predictive economy.

Reference

- Source: PAPER_1307
- Related: PAPER_1494-1635 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "ckm_unitarity_sum_1"})`

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